NEW TECHNOLOGIES RESEARCH CENTRE
UNIVERSITY OF WEST BOHEMIA
IN PILSEN

BIOMECHANICAL HUMAN BODY MODELS

SOFTWARE

Pelvic Floor Morphing

Author(s): Luděk Hynčík, 61950
Adam Wittek, The University of Western Australia

Project number: CZ.02.1.01/0.0/0.0/17_048/0007280
Gledden fellowship, The University of Western Australia

Report number: NTC-ASW-26-001

Leader of department: Jan Vychytil

Director of centre: Petr Kavalíř

PLZEŇ, JUNE 2026

Contents

Mathematical background

The aim of the software module is to register a reference model to a personalized model derived from subject-specific magnetic resonance imaging data.

The reference model is defined by a computational grid (nodes, which form elements). The **registration** is based on the corresponding **anatomical landmarks**, which are significant points on the biological tissues. The subject-specific anatomical landmarks corresponding to those defined on the reference model are acquired from the subject-specific medical images, e.g. magnetic resonance imaging (MRI), computer tomography (CT) or internal and external pelvimetry.

The registration updates the coordinates of the computational grid nodes to subject-specific computational grid nodes to create a personalised model by bringing closer the corresponding anatomical landmarks.

The registration of a reference model to a personalised model consists of 2 consecutive steps, particularly

1. **rigid registration**, which rotates and translates the coordinates of the reference computational grid nodes as close as possible to the position of the subject-specific magnetic resonance images by minimising the distance between the corresponding anatomical landmarks by means of the least squares method, and

2. **morphing**, which deforms the coordinates of the reference computational grid nodes to the position of the subject-specific magnetic resonance images by overlapping the coresponding anatomical landmarks.

Consider the reference model computational grid with n nodes and their coordinates $\mathbf{X}_i = [x_i, y_i, z_i]$, $i \in \{1, \ldots, n\}$ in the matrix form

$$\mathbf{X} = \begin{bmatrix} x_1 & \ldots & x_n \\ y_1 & \ldots & y_n \\ z_1 & \ldots & z_n \end{bmatrix} \tag{1}$$

where $\mathbf{X}$ is a $3 \times n$ matrix. Consider m corresponding landmarks on the reference model and the magnetic resonance images. Let's call the landmarks on the reference model $\mathbf{X}_{Lj} = [x_{Lj}, y_{Lj}, z_{Lj}]$, $j \in \{1, \ldots, m\}$ as the *baseline landmarks*, the reference model as the *baseline model*, the subject-specific landmarks on the magnetic resonance images $\mathbf{X}_{Tj} = [x_{Tj}, y_{Tj}, z_{Tj}]$, $j \in \{1, \ldots, m\}$ as the *target landmarks* and the personalised model as the *target model*.

In the text below, coordinates $\mathbf{X}$ corresponds to the reference model, coordinates $\widehat{\mathbf{X}}$ corresponds to the registered model and coordinates $\overline{\mathbf{X}}$ corresponds to the personalised model.

Rigid registration

The rigid registration rotates and translates the reference model to achieve the closest position to the subject-specific magnetic resonance images by the shortest distance between corresponding anatomical landmarks by means of the least square method. The **Kabsch algorithm** is used for rigid registration.

Kabsch algoritm

The Kabsch algoritm consisting of spatial rotation and translation minimizes root mean square deviation (RMSD). The aim is to find rotation $\mathbf{R}$ and translation $\mathbf{t}$ minimizing

$$\min_{\mathbf{R},\mathbf{t}} \sum_{i=1}^{m} \|\mathbf{R}\mathbf{X}_{Li} + \mathbf{t} - \mathbf{X}_{Ti}\|^2 \tag{2}$$

where $\mathbf{R}$ is the spatial rotation matrix and $\mathbf{t}$ is the spatial translation vector. Firstly the centroids are calculated as

$$\mathbf{P} = \frac{1}{m}\sum_{i=1}^{m} \mathbf{X}_{Li}, \quad \mathbf{Q} = \frac{1}{m}\sum_{i=1}^{m} \mathbf{X}_{Ti} \tag{3}$$

and all landmarks are centralized as

$$\widetilde{\mathbf{X}}_L = \mathbf{X}_L - \mathbf{P}, \quad \widetilde{\mathbf{X}}_T = \mathbf{X}_L - \mathbf{Q}. \tag{4}$$

Using the covariation matrix $\mathbf{H} = \widetilde{\mathbf{X}}_L^\top \widetilde{\mathbf{X}}_T$, the singular value decomposition is done as $\mathbf{H} = \mathbf{U}\boldsymbol{\Sigma}\mathbf{V}^\top$. The optimum rotation is $\mathbf{R} = \mathbf{V}\mathbf{U}^\top$. If $\det(\mathbf{R}) < 0$, the mirroring is corrected as

$$\mathbf{V}[:, -1] \leftarrow -\mathbf{V}[:, -1], \quad \mathbf{R} = \mathbf{V}\mathbf{U}^\top. \tag{5}$$

The translation is calculated as $\mathbf{t} = \mathbf{Q} - \mathbf{R}\mathbf{P}$. Finally, the registered points are calculated as $\widehat{\mathbf{X}}_i = \mathbf{X}_i + \mathbf{t}$, $i \in \{1, \ldots, n\}$, and $\widehat{\mathbf{X}}_{Lj} = \mathbf{X}_{Lj} + \mathbf{t}$, $j \in \{1, \ldots, m\}$.

Morphing

The morphing deforms the rigidly registered reference model (defined by grid nodes coordinates $\widehat{\mathbf{X}}$) to the position of the subject-specific magnetic resonance images to create a personalised model (defined by target computational grid nodes coordinates $\overline{\mathbf{X}}$) by interpolation using **radial basis function**.

Radial basis function

Radial basis functions (RBF) to interpolate function $g(\mathbf{X})$ at the point $\mathbf{X}$ having m landmarks on both the baseline and the target grids takes the form [1]

$$g(\mathbf{X}) = p(\mathbf{X}) + \sum_{j=1}^{m} \lambda_j \varphi\left(\|\mathbf{X} - \mathbf{X}_j\|\right) + \alpha \tag{6}$$

where $p(\mathbf{X})$ is a low order polynomial, λ_j, $j \in \{1, \ldots, m\}$, is the weighting coefficient, φ is the radial basis function, $\|\mathbf{X} - \mathbf{X}_j\| = \sqrt{(x - x_j)^2 + (y - y_j)^2 + (z - z_j)^2}$ and α is a constant. The first order polynomial selected for $p(\mathbf{X}) = [1, x, y, z]\,[c_0, c_1, c_2, c_3]^{\mathrm{T}} = \mathbf{Xc}$ and φ is the radial basis function.

Substituting the rigidly registered baseline landmarks $[\widehat{x}_{Lj}, \widehat{y}_{Lj}, \widehat{z}_{Lj}] = \widehat{\mathbf{X}}_{Lj}$, $j \in \{1, \ldots, m\}$ and the target landmarks $f(\mathbf{X}) = [x_{Ti}, y_{Ti}, z_{Ti}] = \mathbf{X}_{Ti}$, $j \in \{1, \ldots, m\}$ to Equation (6) brings

$$\mathbf{X}_{Ti} = \sum_{j=1}^{m} \lambda_j \varphi\left(\|\widehat{\mathbf{X}}_{Li} - \mathbf{X}_{Tj}\|\right) + \alpha + \widehat{\mathbf{X}}_{Li}\mathbf{c}, \quad j \in \{1, \ldots, m\} \tag{7}$$

or

$$\begin{bmatrix} \mathbf{A} + \alpha\mathbf{I} & \mathbf{B} \\ \mathbf{B}^{\mathrm{T}} & \mathbf{0} \end{bmatrix} \begin{bmatrix} \boldsymbol{\lambda} \\ \mathbf{c} \end{bmatrix} = \begin{bmatrix} \mathbf{T} \\ \mathbf{0} \end{bmatrix} \tag{8}$$

in the matrix form, where matrix $\mathbf{A} = A_{ij} = \varphi\left(r_{ij}\right)$ is a square $m \times m$ matrix containing the distances between the mutual pairs of the baseline landmarks $r_{ij} = \|\widehat{\mathbf{X}}_{Li} - \widehat{\mathbf{X}}_{Lj}\|$, $\forall i, j \in \{1, \ldots, m\}$, $\mathbf{I}$ is the identity matrix of size $m \times m$,

$$\mathbf{B} = \begin{bmatrix} 1 & \widehat{x}_{L1} & \widehat{y}_{L1} & \widehat{z}_{L1} \\ \vdots & \vdots & \vdots & \vdots \\ 1 & \widehat{x}_{Lm} & \widehat{y}_{Lm} & \widehat{z}_{Lm} \end{bmatrix} \tag{9}$$

is a $m \times 4$ matrix containing the coordinates of the baseline landmarks,

$$\mathbf{T} = \begin{bmatrix} x_{T1} & \ldots & x_{Tm} \\ y_{T1} & \ldots & y_{Tm} \\ z_{T1} & \ldots & z_{Tm} \end{bmatrix} \tag{10}$$

is a $3 \times m$ matrix containing the coordinates of the target landmarks and $\mathbf{c}$ is constant matrix. Solving Equation (8) brings the matrix $\boldsymbol{\lambda}$ of size $m \times 3$ and the matrix $\mathbf{c}$ of size 4×3. Once they are determined, by assuming that the number of nodes in the baseline grid is n, the coordinates of the nodes in morphed grid $\overline{\mathbf{X}}$ (a matrix of size $n \times 3$) can be calculated as

$$\begin{bmatrix} \overline{\mathbf{X}} \\ \mathbf{0} \end{bmatrix} = \begin{bmatrix} \overline{\mathbf{A}} & \overline{\mathbf{B}} \\ \mathbf{B}^{\mathrm{T}} & \mathbf{0} \end{bmatrix} \begin{bmatrix} \boldsymbol{\lambda} \\ \mathbf{c} \end{bmatrix} \tag{11}$$

where

$$\overline{\mathbf{X}} = \begin{bmatrix} \overline{x}_1 & \overline{y}_1 & \overline{z}_1 \\ \vdots & \vdots & \vdots \\ \overline{x}_n & \overline{y}_n & \overline{z}_n \end{bmatrix} \tag{12}$$

is a $n \times 3$ matrix containing the coordinates of the nodes in the morphed grid, matrix $\overline{\mathbf{A}} = \overline{A}_{ij} = \varphi(r_{ij})$ is a $n \times m$ matrix, where $i \in \{1, \ldots, n\}$ are the nodes in the baseline grid, $j \in \{1, \ldots, m\}$ are the baseline landmarks and

$$\overline{\mathbf{B}} = \begin{bmatrix} 1 & \widehat{x}_1 & \widehat{y}_1 & \widehat{z}_1 \\ \vdots & \vdots & \vdots & \vdots \\ 1 & \widehat{x}_n & \widehat{y}_n & \widehat{z}_n \end{bmatrix} \tag{13}$$

is a $n \times 4$ matrix containing the coordinates of the nodes in the baseline grid. The radial basis function can be chosen either

- **thin plate spline** [3], which has a global support and is defined as

$$\varphi(r) = r^2 \log r \tag{14}$$

  where $r = \|\mathbf{X} - \mathbf{X}_i\|$, which generally results in a smooth mesh [2], or

- **Wendland C2 compact support function** [4], which has a local support, is C^2-smooth and is useful for morphing in biomechanics. It is defined as

$$\varphi(r) = \begin{cases} (1-r)^4 (4r+1), & 0 \le r \le 1, \\ 0, & r > 1, \end{cases} \tag{15}$$

  where $r = \frac{\|\mathbf{X} - \mathbf{X}_i\|}{R}$ and R is the support radius.

Implementation

The software module concerns the 2026 version of 3D Slicer extension *Pelvic Floor model* (PelFoM) available at GitHub, see Figure 1.

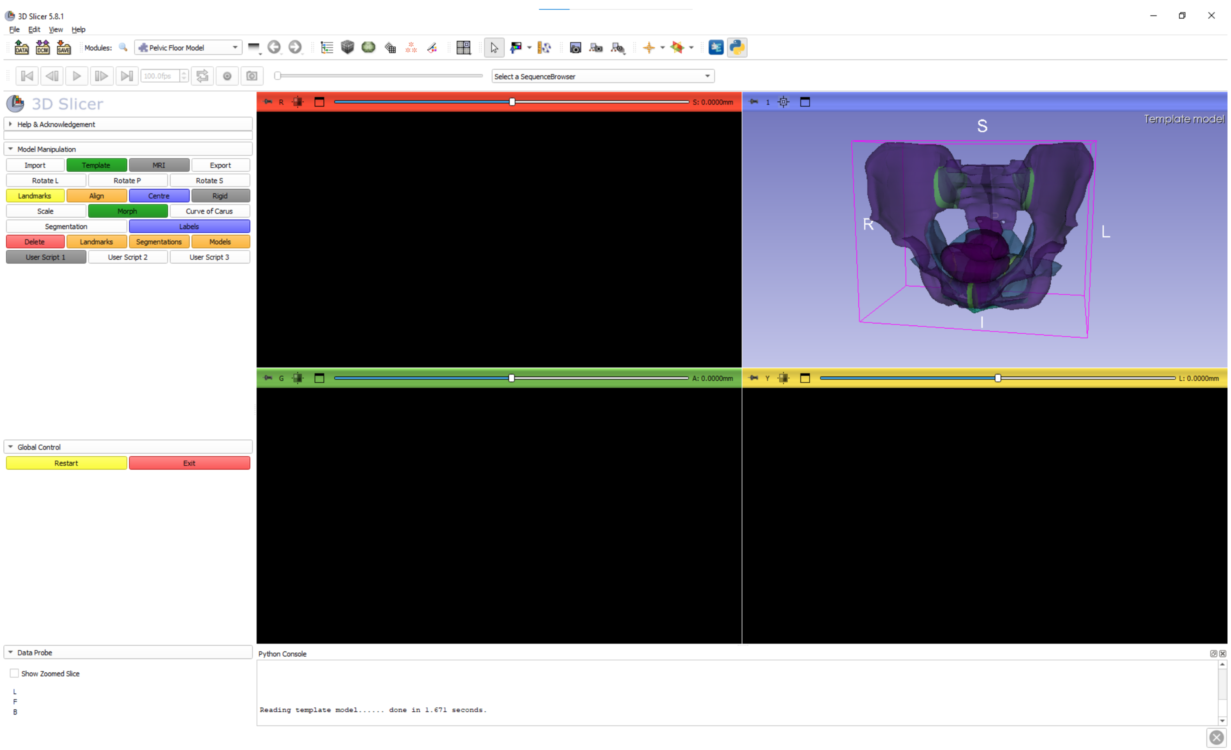

Figure 1: Environment eith the reference model

License

The module is available under a BSD 3-Clause "New" or "Revised" License.

Installation

For installation, follow the steps in the Installation Manual. Developers may use Developers Manual and Code Structure. The mathematical theory of registration is described here and available online.

Reference model

The module consists of a reference bony pelvis model (see Figure 1) including pre-defined anatomical landmarks (see Table 1 and Figure 2).

Anatomical landmarks

Landmark	**Anatomical description**
PSA	Anterior point of the pubic symphysis (superior border)
PSP	Posterior point of the pubic symphysis
PSI	Inferior point of the pubic symphysis (inferior border)
L5	Spinous process of the fifth lumbar vertebra
PR	Sacral promontory
S3	Midpoint of the superior surface of the third sacral vertebra (S3)
SJ	Sacrococcygeal joint (internal landmark)
SB	Sacrococcygeal joint (external surface landmark)
ILL	Left-most point on the iliopectineal line
ILR	Right-most point on the iliopectineal line
BCL	Left iliac crest (distantia bicristalis)
BCR	Right iliac crest (distantia bicristalis)
BSL	Left anterior superior iliac spine (distantia bispinalis)
BSR	Right anterior superior iliac spine (distantia bispinalis)
ISL	Left ischial spine
ISR	Right ischial spine
ITL	Left ischial tuberosity
ITR	Right ischial tuberosity
TIL	Left tuber ischiadicum
TIR	Right tuber ischiadicum
PSISL	Left posterior superior iliac spine (skin dimple)
PSISR	Right posterior superior iliac spine (skin dimple)
TRL	Left greater trochanter
TRR	Right greater trochanter

Table 1: Anatomical landmarks

Acknowledgement

The work was supported by the project "Integration of biomedical research and health care in the Pilsen metropolitan area“; reg. no. CZ.02.01.01/00/23_021/0008828) - co-funded by the European Union and by the State Budget of the Czech Republic.

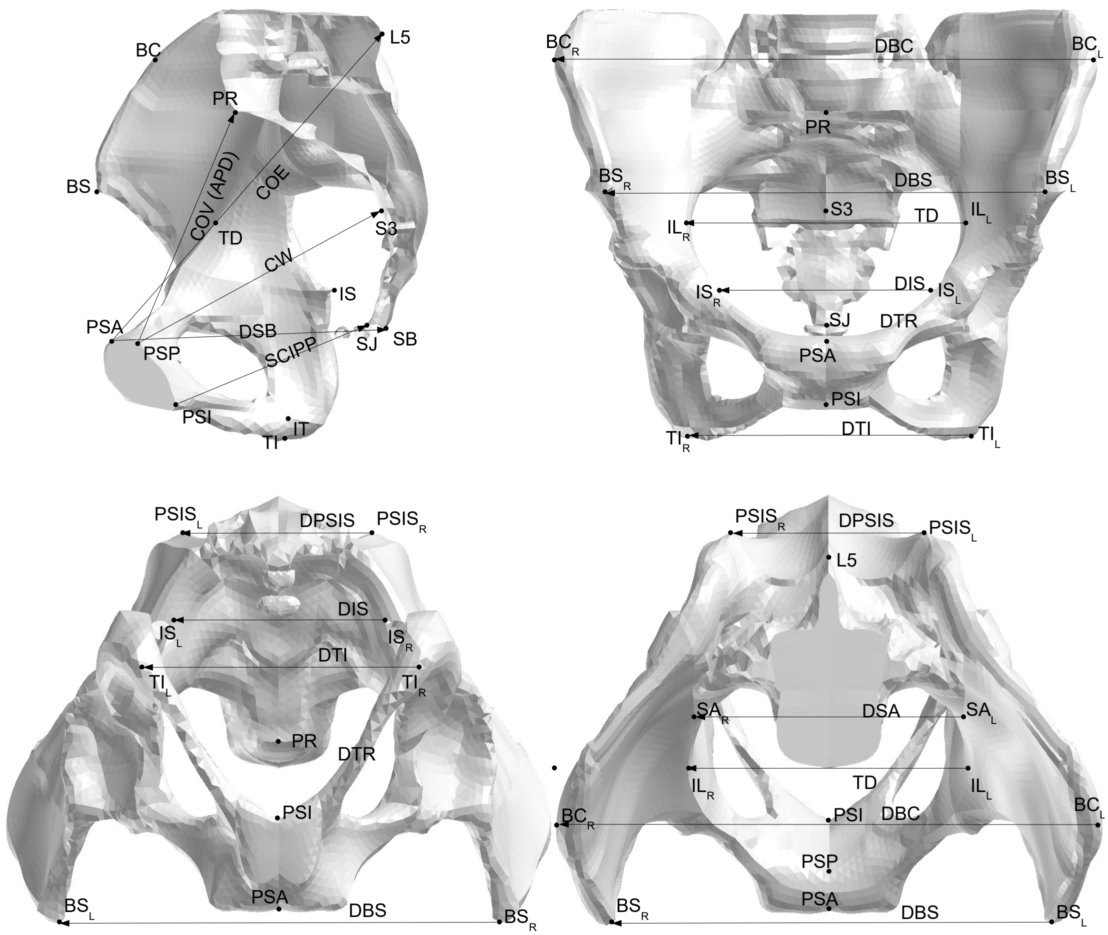

Figure 2: Anatomical landmarks

References

[1] J. C. Carr, R. Fright, and R. K. Beatson. Surface interpolation with radial basis functions for medical imaging. *IEEE Transactions on Medical Imaging*, 16(1):96–107, 1997.

[2] E. Hwang, J. Hallman, K. Klein, J. Rupp, and al. Rapid development of diverse human body models for crash simulations through mesh morphing. *SAE Technical Paper 2016-01-1491*, 2016.

[3] V. Skala. A practical use of radial basis functions interpolation and approximation. *Investigacion Operacional*, 37(2):1–8, 2016.

[4] H. Wendland. Piecewise polynomial, positive definite and compactly supported radial functions of minimal degree. *Advances in Computational Mathematics*, 4:389–396, 1995.